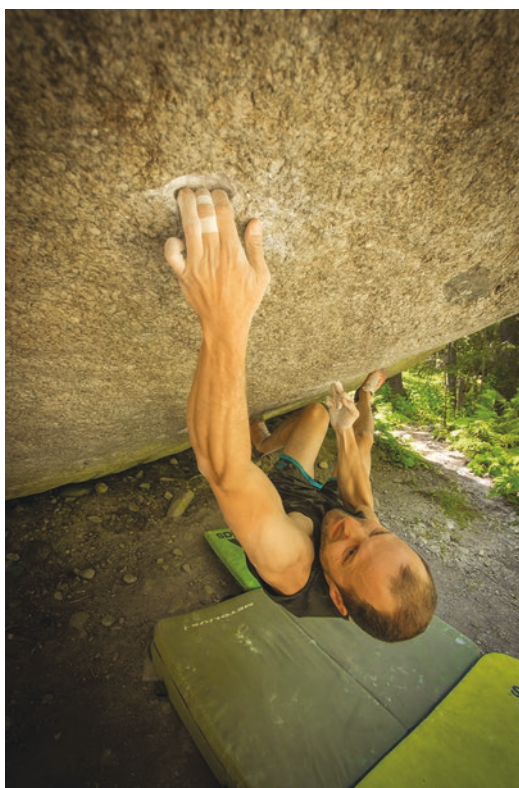


Sport Climbing with Pre-existing Medical Conditions

17

T. Küpper and A. Morrison



Christoph Lutter bouldering at Charmonix, France, Photo Michael Simon

T. Küpper (✉)

Institute of Occupational, Social and Environmental
Medicine, RWTH Aachen Technical University,
Aachen, Germany
e-mail: tkuepper@ukaachen.de

A. Morrison

Royal Free London NHS Foundation Trust,
London, UK

Pre-existing medical conditions may—but not always—set limitations on physical exercise and sporting participation. Some adjustments in frequency, intensity, time, and type (FITT) may be required, including to sport climbing. The

most important message first—it is very rare that patients must stop climbing completely!

There is a broad consensus that regular physical activity and exercise are an essential part of a healthy lifestyle with significant health benefits and are regularly prescribed for the prevention and treatment of many diseases [1]. For example, aerobic exercise and aerobic with resistance exercise can significantly improve the aerobic capacity of chronic heart failure patients, whereas resistance exercise alone cannot [2]. Other meta-analyses confirm that both aerobic and dynamic resistance training can reduce hypertension [1, 3], and small studies show climbing can reduce backache [4], even depression [5].

Limb (1995) reported no cardiac events at UK indoor climbing walls with over 1.7 million users. All exercise adapts the heart accordingly. A trained sport climber's heart is characterized by a submaximal cardiac output, high to maximal heart rate, and moderate to marked increase in peripheral resistance [6]. This is explored later, but sport climbing is essentially a mix of aerobic plus resistance exercise combined with significant hand-gripping, where performance limiting or terminating "climbing fatigue" is not systemic but generally localized to the forearm flexor muscles responsible for gripping. It's unlike standard cycling or running treadmill exercise testing where performance is predominately aerobic and terminated by breathlessness and fatigue of the exercising large muscle groups. Both standard ergometric exercise tests can also prescribe a steady exercise state for the heart rate (HR), which is impossible in climbing. Climbing's unique physiological response to each route combines submaximal aerobic and anaerobic pathways, and a disproportionate rise in HR and blood pressure (BP), especially when climbing nearer the maximal grades of climber [6].

There is no standardized method to test climbing exercise. However, Sheel (2004) compared cycling ergometry VO_2 to the VO_2 when climbing on grades 2 and 3 below the maximum grade of the climber [7, 8]. Sheel found the respective climbing grade's VO_2 to be 45 and 51% of the cycling VO_2 max.

In assessing over one million standard ergometry exercise tests, there were no fatal or nonfatal complications in the athletes, and

in the remaining 712,285 predominantly with coronary heart disease, there were 17 deaths and 96 life-threatening complications, most frequently regarding ventricular fibrillation [9]. Gibbons et al. conducted 71,914 maximal exercise tests with uniform conditions in a single medical facility in a population with low prevalence of known heart disease from 1971 to 1987 [10]. No complications occurred from 1977 in 45,000 maximal tests. The overall cardiac complication rate in women and men is 0.8 complications per 10,000 tests with 95% confidence intervals of 0.3–1.9 complications per 10,000 tests. Standardized maximal exercise testing was found to be safer than previously published, but there still remains a very minor risk to be addressed. Fletcher et al. clearly defines various parameters for ergometry testing in those with cardiovascular disease, with epidemiological data demonstrating the necessity of regular exercise in these patients, and the resulting positive health benefits [11].

Climbers are generally dedicated to the sport, be it recreational or competitive. They may have long walks with rucksacks to outdoor crags and/or regularly participate in other exercise which may include cycling to a climbing gym, strength training, or other sports participation. However, whenever there are pre-existing medical conditions, the climber must seek the advice of a doctor who is well informed about the respective diagnosis in sport medicine and who is preferably also experienced in climbing. Sport climbing's physiology is often misunderstood or dismissed as there's limited data published with respect to exercise prescription. The patient should not be unnecessarily limited in their regular activities as the benefits of regular exercise and sports are significant and almost exclusively outweigh the risks.

There must always be regard for any disease process itself, and each climber presents with a different FITT training background. They may require a baseline or maximal level of fitness to be established and/or comprehensive history reported.

In seeking the best possible medical advice, the following questions must be answered:

1. Is there any risk that the disease may be worsened by sport climbing?
2. Alternatively, does the disease limit the ability to climb?
3. Does the disease cause any risk while climbing?
 - (a) When leading?
 - (b) When belaying?
4. Are there any additional risk/s from environmental factors which may worsen the disease?

The following are examples of replies to the four questions raised above.

1. Diabetics should monitor their blood sugar during and after sport as exercise stresses glucose metabolism.
2. Limited mobility in the case of rheumatic or orthopedic diseases may cause limitations when climbing.
3. Seizures may be a risk when leading or belaying but not when seconding or top rope climbing;
4. If a diabetic has peripheral angiopathy or a climber suffers from Raynaud's disease or other vascular dysfunction, cold weather may cause significant problems and may increase the risk of frostbite.

For further clarification, the following questions should also be asked:

1. Which—if any—may be a limiting factor to climb?
2. Is the disease in an acute phase, or has it been stabilized?

Based on these answers and—if necessary—additional clinical tests, the individual recommendation should be given. Again, in most cases, the patient will be able to continue climbing if he/she accepts some rules. Principally, any acute or non-stabilized situation should be stabilized by adequate therapeutic procedures before any assessment of fitness to climb is answered. It should be noted that there are nearly no evidence-based recommendations and

data specifically about climbing with pre-existing diseases [6], but there are for related sports. Therefore, the bulk of the recommendations, also those from international panels—e.g., the UIAA Medical Commission (Union Internationale des Associations d'Alpinisme)—are based on a consensus of international experts who are climbers themselves and who transfer sports medical knowledge from other disciplines to sport climbing [12–16]. It should be noted that most of these recommendations focus on “classic” alpine climbing and not sport climbing.

The following recommendations for sport climbing with pre-existing medical conditions will consider each condition in turn. However, some physiological understanding of the sport will first be discussed to avoid further repetition in the respective diagnoses that follow.

17.1 Physiology of Sport Climbing Seen with the Eyes of Exercise Physiology

Climbing is often said to be “strenuous” without any understanding or qualification of what this means. How fit must a person with pre-existing disease be to climb without taking inadequate risk for himself or fellow climbers? Answering this requires an understanding of baseline physiological parameters induced by sport climbing. Sport climbing is an intermittent activity where each unique route has a graded difficulty level once climbed, and this is typically followed by a short rest or belaying one's partner. It involves complex static isometric and dynamic moves.

Higher grades become incrementally more strenuous, as are routes with increasing incremental overhangs angles or “roof” routes. The lowest grade is equivalent to a scramble walk. Trained climbers can climb more routes of higher grades in a single session because of their enhanced training status; and if more experienced, they have additionally honed efficient climbing techniques to reduce strenuous loading (i.e., not over gripping and thus delaying the onset on climbing-specific fatigue, better footwork, and/or core strength). It is well established

that climbing increases heart rate (HR) disproportionately relative to the “exercise” of climbing and can fluctuate between 129 and 180 bpm during a climb, with peaks greatest on increasing inclines on overhanging routes [6].

Accordingly, a trained sport climber’s heart is characterized by a *submaximal cardiac output, high to maximal heart rate, and moderate to marked increase in peripheral resistance* [6]. As the sport is not predominately aerobic, but also involving resistance training, heart rate and blood pressure will be disproportionately high during the climb itself, with irregular breathing, which return to normative values soon after the climb [6]. A typical climb lasts 2–7 min, with blood lactates remaining elevated over 20 min post-climb [17]. Sport climbing does not require a high level of cardiovascular fitness with an adult $\text{VO}_{2\text{peak}}$ during the climb estimated to be 52–55 mL/kg⁻¹/min⁻¹ for maximum oxygen uptake, averaging 20–25 mL/kg⁻¹/min⁻¹. The latter is equivalent to the stress tolerance in an untrained population [17]. The $\text{VO}_{2\text{max}}$ results for sedentary people undergoing isometric exercises are <45 mL/kg⁻¹/min⁻¹, while elite cyclists have >68 mL/kg⁻¹/min⁻¹. Therefore, measurements of the maximal oxygen uptake ($\text{VO}_{2\text{max}}$) are not representative for the real strain of sport climbing [18].

In a study investigating whether there was any advantage in an active pause when sport climbing, the probands who climbed a 20 m route with a crux of VIII+ (UIAA scale) showed a mean heart rate (HR) of 148 bpm (+/-16), a mean oxygen uptake (VO_2) of 1660 mL/min (+/-340), and a relative oxygen uptake of 24.7 mL/kg⁻¹/min⁻¹ (+/-5.3) [18]. Short peak loads require 31.9 mL/kg⁻¹/min⁻¹ (+/-5.3). An active pause had no advantage over a passive one. However, it must be criticized that lactate was not investigated although it is well known that its increase requires a significantly longer recovery. With a passive pause, lactate is still increased for 30 min post-climb before getting back to normal levels or continued increasing for 20 min after an active pause [18].

Given this physiological understanding, it can be concluded that sport climbing makes no significant demands on general endurance nor

does it require a high cardiovascular fitness for heart patients; the data suggests sport climbing exercise corresponds to loads which are recommended as secondary prevention in heart cases.

As expected, climbing routes closer the individual’s grade limit will cause a significantly higher peaks in HR and higher VO_2 [6]. However, when climbing moderate routes, HR is at 67%, and VO_2 is 45% of the individual’s maximum VO_2 for cycling ergometry [8]. This again indicates that standard exercise ergometry tests should not be used for assessing climbing fitness because the (largely) linear correlation between HR and VO_2 does not exist during climbing. Such correlation will be found in aerobic endurance disciplines where the body’s large muscle groups become fatigued and the subject breathless, unlike climbing where performance limiting fatigue is generally localized to hand gripping function. In consequence, the measurement of the ischemic threshold of a patient with coronary heart disease may be used, but the VO_2 of a pulmonary patient is irrelevant if this patient climbs.

Some authors also mention the psychological factors that may additionally impact on the disproportionality between HR and VO_2 when climbing, e.g., stress and fear [8, 19, 20]. Williams et al. examined the heart’s response to climbing outdoors with experienced and novice climbers using ECG and placebo or beta blockade to limit HR peaks [20]. Williams concluded, “...*this popular sport represents more an anxiety-type of psychological stress than a physical stress and as such is likely to increase moral fibre rather than muscle fibre.*” However, the testing day was wet, route slippery and insecure, especially for novices. Experienced outdoor climbers will generally deal with inclement weather and bold movements much better than novices. Beta blockade does not improve climbing performance, though incorrectly thought to be important to climbing performance among some climbers in the 1970s. Although studies are few, experienced climbers generally manage the “fear” of a bold move much better than novices, irrespective of weather, and have lower HRs; and more significantly, experienced climbers also don’t overgrip holds which will raise HR [6]. Perhaps it is not surprising that

climbing training is relevant for 59% of the performance only [21].

Interestingly, the anaerobic strain is much higher when comparing indoor climbing to outdoor. While the HR, VO_2 , and lactate at rest were identical, the maximal lactate was significantly higher (10.4 mmol/L) for indoor climbers compared to climbing natural rocks of similar difficulty [22]. All studies demonstrate that lactate is also not a valid parameter to measure fitness of climbers as there are no steady states in exercise to be achieved while climbing, and each graded route differs in its “exercise” intensity. This may be different when considering local lactate, especially if the arms would be measured, but this testing is not realistic for normal climbing.

Although there are several limitations when applying results from traditional exercise testing protocols (cycling, treadmill) to sport climbing, some important conclusions may be drawn for climbers with pre-existing conditions [19, 23]. With an increase of 54–85% of HR_{max} and a VO_2 which corresponds to easy jogging, climbing may be recommended as discipline for secondary prevention, e.g., for diabetes or coronary heart disease.

17.2 Acute Infections

The most frequently encountered and probable cause acutely affecting the heart is acute febrile infection. This is assumed to be an important factor for sudden and severe complications or even deaths in sports via myocarditis as third frequent cause of death [24], and arrhythmia [25–28]. Although there are no valid data available, obviously different viruses behave differently, and those that are “critical” are Parvovirus B19, herpes viruses, and entero- and adenoviruses [29–31]. When outdoor climbing, it should be taken into account that in several regions a myocarditis may be a complication of a borreliosis [30].

Independent from the sporting discipline, there is international consensus in sport medicine and occupational medicine that demanding activities should not be performed at least for twice the number of days that the patient had

fever. Non-demanding activities like compensatory training is possible if the patient feels well enough to do so.

17.3 Sport Climbing with Diabetes Mellitus

Summary

- **Limiting factor:** Endurance, not maximal force, climbing grade difficulty level, characteristics of the route, altitude, etc.
- **Additional risk factors:** Cold, heat, minor trauma, (shoes too tight!)
- **Common mistake:** Blood sugar measured before and while climbing and also some hours after exercise

There are no studies available concerning sport climbing and diabetes. However, there is international consensus in sport medicine that regular sport is a good secondary prophylaxis for diabetics and especially for type 2 diabetes; this is supported with good data. The main effect is the build-up of metabolic highly active muscle tissue.

If there is no significant neuropathy, the maximal force and short-time-endurance of diabetics is not reduced, and, vice versa, side glucose metabolism will not be influenced significantly by the activity. Therefore, the risk for hypoglycemia is reduced compared to endurance sports, e.g., mountaineering. Hence, such complications are extremely rare in climbing gyms (Küpper, unpublished data). However, consequent monitoring of the blood sugar should be integral part of any sportive activity of diabetics with another measurement some hours after climbing included. Often patients will forget the latter when they climb in the evening, and the test should be performed then after midnight. Overall, blood sugar management is much easier now with modern equipment which is small and lightweight, some of which offer automatic and/or continuous measurement. So far, no contraindications were found for such devices, but the user should always keep up-to-date with the respective specifics of the different systems.

Basically there are continuous glucose monitoring systems (CGM) and flash glucose monitoring systems (FGM). CGMs measure the sugar in the liquid of the subcutaneous tissue more or less continuously, at least in short intervals. For FCMs, the sensor is located also in the subcutaneous tissue in short intervals, but the results will be stored on a microchip and then read by a special device or via smartphone. Data from both systems may be stored and evaluated digitally and used later for advice by the sports medicine specialist or the diabetologist. For both systems, a very small platinum electrode must be placed into the subcutaneous fat. It doesn't disturb the patient, but dependent on the system, it must be replaced every few weeks. Independent unsystematic observations from a limited number of cases have shown that the systems are well tolerated by the climbers. The location of the sensor at the dorsal upper arm generally does not constrict. This would be the case only for a very limited number of moves, e.g., back and footing, where the arm is used backward.

CGM and FGM must be differentiated from noninvasive continuous measurement (NCM). The latter is of advantage for persons with phobia for needles. No device must penetrate the skin. However, this system has a significant and unratable disadvantage. NCM is based on the reflexion of laser light, which points at the skin's sweat. Here, several substances are solved (glucose, protein, water, uric acid, etc.), and their collective measurement allows a conclusion of blood sugar concentration. For people at rest, the results are of acceptable validity. However, during sports, the composition of sweat differs significantly from those at rest. Another problem is that in people who are used to sweating regularly (sports, sauna, etc.), sweat is produced more effectively than in other situations which may mean their sweat contains much more water and less of other (valuable) substances, especially salt. Research demonstrated that this effect cannot be taken into account by available systems thus far. Seen from the eyes of sport medicine, such systems cannot be generally recommended in contrast to CGM and FGM systems.

The necessary validity and exactness of measurement has been included into international standards. Since 2016, only those devices meeting the testing criteria set out in ISO 15197 can be sold. With an accepted tolerance of $\pm 15\%$, CGM and FGM systems fulfill this standard, but NCM show an error of measurement of $\pm 30\%$ or more which is not acceptable according to ISO 15197. The exactness results of watches or wristband fail to meet these criteria. These systems will give an alarm to the user when the limits should be exceeded, but this is not reliable enough because of data variance. However, these systems may be used as backup for others. The exactness of measurement must be taken into account also when the "traditional" test with strips should be used. Such tests are linked chemical-optic tests and are therefore temperature dependent. When used below $+14\text{ }^{\circ}\text{C}$, the reported blood sugar concentration is false-low (below $0\text{ }^{\circ}\text{C}$ any measurement is impossible). When the temperature should be higher than $+35\text{ }^{\circ}\text{C}$, the result will be a false high. Therefore, these devices should be carried under warmth and shade of jackets on cold days, and similarly in the shade during hot days. As the measurement of these test substance devices are UV light sensitive, ensure the container with the testing strips is opened for as short a time as possible and in the shade.

Checkup for Climbing Diabetics

- **History:** Experience in climbing, training status, nutrition, blood sugar management, hypoglycemic phases in the past (especially when correlated to sports), medication, knowledge of individual symptoms for hypoglycemia
- **Actual status:** Vessels (peripheral angiopathy, especially of the fingers or feet?), neurology (peripheral neuropathy?), ocular fundus
- **Laboratory:** Creatinine, urine status, HbA1c
- **Technical investigations:** ECG at rest, ergometry
- **In the case of individual indication:** Stress echocardiography, 24 h ECG, ultrasound of the kidneys, etc.

A precondition for diabetics to undertake safe sporting participation is the continued knowledge

about the disease, good compliance to improve diabetes control, and the absence of any secondary organ damage which may cause limitations. The latter would necessitate special advice with a careful risk and benefit assessment included.

Independent from the discipline, fellow climbers should be informed of the diabetic's diagnosis and first aid involving the following options should they arise. They should also know where the diabetic carries some carbohydrates (normally readily accessible in the top compartment of the rucksack) [16]. Any emergency kit should include some carbohydrates. On the strength of past experiences using cereal bars as the carbohydrate, often recommended especially in Anglo-American literature, this is not the best choice. They are bone-dry and make the athlete thirsty. Much better are carbohydrate gels which are state of the art for diving diabetics. They can be consumed directly from the packet, and thirsty persons tolerate them much better. To carry along glucagon or insulin in the emergency bag is a topic of discussion. If climbers do carry these, they must be confident in using them correctly.

The limited temperature tolerance of diabetic drugs should also be taken into account as they are quite resistant against heat but should never be allowed to freeze [32]. Only about 15% of the normal population knows the local emergency number. Therefore, this should be included in the emergency kit as the number can vary. For example, in most countries worldwide, 112 is the emergency calling number, but its 911 in North America. In some African and Asian countries, there are other local numbers.

With respect to tactics for managing blood sugar levels, the following points should be noted. After a strenuous day, the blood sugar level may be a little higher than normal in the following morning, and this is fine. When significant endurance exercise is anticipated in a given day, the amount of carbohydrates consumed at breakfast should be increased by about 30%. In hot climates, the hydration status should be monitored carefully. Dehydration is more dangerous for diabetics than a mild hyperhydration, and therefore intake should be liberal [16]. Special attention

should be given to the treatment and prevention of injuries, also to minor wounds [16].

Even with good compliance, diabetics will develop some minor microangiopathy and neuropathy in their daily routine which is irrelevant. In cold weather, this increases the risk of frostbite as the diabetic person will not feel the effects of cold as much as a healthy person due to the neuropathy; and if frostbite has already occurred, the healing will be more inferior for the diabetic due to the microangiopathy. Protection from the cold should be actioned early by wearing non-constrictive shoes and gloves and, if necessary, fingerless gloves for climbing. Perfectly fitted shoes and regular care for the nails and feet are a must for diabetics [16]. Never wear shoes sized too small. Comfort is the measure of all things! As soon as pressure marks occur from ill-fitting footwear, climbing must be stopped immediately, and it should be considered whether to buy larger or better fitting shoes. If wounds should occur at the hands or feet, however minor, these should be treated carefully. This includes perfect disinfection, covering, and no climbing until fully healed.

17.4 Sport Climbing with Cardiovascular Disease

For most cardiovascular diseases, there is no doubt that exercise is an important factor for the prescription of secondary prevention [2, 3, 33–35]. Therefore, it must be strongly warned to advise a climber to stop climbing without significant arguments or not to continue climbing with specific advice by a specialist.

17.4.1 Hypertension

One may call hypertension a pandemic of modern civilization, and, of course, it also affects climbers. Some sport medicine specialists unfamiliar with climbing physiology give the undifferentiated advice to “stop climbing” when presented with climber with hypertension. It is conspicuous that the former often “never in their life have climbed higher than a bar stool” (cit. Ulf Gieseler,

Heidelberg/Germany). They should consider the following argument. If assessed from an exercise physiology viewpoint, a hypertensive person, who is treated perfectly, who does not suffer from secondary damages of hypertension, and who shows a normal blood pressure regulation at rest and during exercise, should be viewed as a healthy person able to climb [6]. Meta-analyses confirm that both aerobic and dynamic resistance training can also reduce hypertension, among many other health benefits to be derived from exercise, and are endorsed by multiple international hypertensive consensus recommendations [1, 3].

Summary

- **Limiting factors:** Maximal force, especially when combined with a whole-body load and forced respiration during expulsion (and therefore indirectly the difficulty of the climb), may cause risk but not the length of the climb, endurance exercise, altitude, climate, etc.
- **Additional risk factor:** Competition situation, “pushing the limits”
- **Common mistakes:** Compensation of a lack of technique with a more of power, climbing of significant overhanging routes only, not adequate pauses between the climbs

Also for hypertension, it should be noted that data are insufficient. However, some observations indicate strongly that well stabilized hypertensive patients without secondary organ damages show a normal exercise-adequate increase and decrease of their blood pressure [6, 36]. The specific problem is that the method of Riva-Rocci cannot be used during climbing, and for technical reasons, the earliest measurement will only be possible at least 1 min after the climb when blood pressure has already decreased. A study by the authors with continuous blood pressure measurement during climbing has been approved by the ethical commission but has not yet commenced.

There is one study which tried to evaluate whether the climbing of a hypertensive patient might cause a risk for the myocardium. It is unfortunate that the study design was weak, but no effect was found in 19 participants [37].

However, although data are scarce, some conclusions may be drawn to advise hypertensive climbers. Experienced climbers showed a systolic blood pressure increase of about 13 mmHg, while novices showed even more [38]. From this, it may be concluded that good climbing technique is always advantageous to develop and learn, especially for hypertensive climbers. Near the individual limit, blood pressure will increase disproportionately during the climb and reduce post-climb [6, 36].

Checkup for a Hypertensive Climber

- **History:** Compliance? Check blood pressure recordings! Experience in climbing? Exercise-related symptoms? Secondary organ damage?
- **Actual status:** Blood pressure, body weight, heart, lungs, vessels, ocular fundus
- **Laboratory:** Creatinine, urine status
- **Technical investigations:** ECG at rest, ergometry, 24 h blood pressure
- **In special situations/individual indication:** Echocardiography (stress), thorax X-ray, ultrasound of the kidneys, etc.

There is no climbing limitation when blood pressure has been stabilized within a physiological range, the diurnal rhythm is intact, blood pressure response is adequate to the respective exercise, and there are no secondary damages of organs (these might define a limitation) [6].

When the systolic blood pressure is higher than 160 mmHg, or diastolic above 100 mmHg, the situation should be handled with care. The climber may continue climbing but at a lower level until an adequate therapy has been established. Overhanging routes should be strictly avoided for this period. However, if the pressure should be higher than 200 mmHg systolic and above 120 mmHg diastolic, any sport should be stopped until sufficient therapy has lowered such pressure into an acceptable range. Then, secondary damage of organs should be excluded before the continuation of climbing may be discussed [6]. If such damage should be identified, it should be considered whether they cause a limitation for climbing, or vice versa, whether climbing may worsen the damage. In most cases, it will be pos-

sible to continue climbing at a “level of joy” without competition and “pushing the limits.” Routes with significant overhangs should be avoided.

Although there are no data so far, a careful view on the characteristics of the respective climb should be recommended. Maximal loads near or at the individual limit and therefore with and forced respiration during expulsion should be avoided. This would be especially the case in hard overhangs or roofs. Counterforce techniques (“Piazen”) would have a similar effect, although less pronounced. In contrast, slabs and walls would be beneficial.

For optimal therapy, there are international recommendations which will be regularly updated. Because maximal aerobic endurance exercise and maximal heart rate (HR) do not really play a significant role in sport climbing, the decrease of HR_{max} and VO_{2max} by beta-blockers can be neglected. However, a careful follow-up of any update of the World Anti-Doping Agency (WADA) regulations should be provided whether the ban of such drugs—actually limited to some disciplines, e.g., all shooting—may be expanded to climbing. Here beta-blockers do not decrease performance but increase it probably by a decrease of nervousness and anxiety. Then, the hypertensive climber with beta-blockers for therapeutic use needs a TUE certificate (“therapeutic use exception”) if a competition should be planned, the same for diuretic drugs if they are part of the therapy. All other antihypertensive drugs are so far not critically concerning anti-doping rules.

17.4.2 Coronary Heart Disease (CHD)

There is an increase of cardiac emergencies with increasing age, also in active or sporting people independent from the discipline, although less than in sedentary persons (survey in [6]). CHD is one of the most important underlying diseases. It is less known that the positive predictive value of a single ergometry of an asymptomatic person is 55–65% only and increases only after several ergometries [39–41]. Gibbons et al. found the overall cardiac complication rate for cycling ergometry in healthy adults to be 0.8 complica-

tions per 10,000 tests, though later the complications were nil for many years [10]. Scherer and Kaltenbach used ergometry testing in 712,285 patients with predominantly CHD, and there were 17 deaths and 96 life-threatening complications, most frequently regarding ventricular fibrillation [9]. The safety of various ergometry testing for this population has been refined and aids personalized exercise prescription and significantly improves the health of such patients [11].

Based on this data, most authors recommend to check persons who start with any sport at the age of about 40+, but a general screening of the symptom-free population doesn’t make any sense [41]. Consequently, this is also not included in most national or international guidelines.

Summary

- **Limiting factor:** Maximal workload (myocardial) but not maximal workload of peripheral muscles like arms, hands, or legs. Length of the climb, especially when it is a hard one. Altitude (hypoxia), climate, and others
- **Additional risk factors:** Competition, “pushing the limits”
- **Common mistakes:** Inadequate technique will be compensated for by more muscular strength or an abrupt increase of climbing difficulty—both causing forced respiration during expulsion.

However, the above recommendations need a modifying statement to calm these down. Thousands of elderly climbers, including those with CHD, visit climbing gyms regularly, and cardiac emergencies in sport climbing are extremely rare events. Limb reported no cardiac events at UK indoor climbing walls with over 1.7 million users [42]. It should be reminded that the general exercise tolerance does not reflect the physiological situation when climbing. However, this is fortunately not exactly the problem for the climbing CHD patient. Such patients don’t need data concerning their maximum workload; they need their so-called ischemic threshold. This is the pulse rate at the workload when first symptoms of CHD should occur or first typical changes of the ECG curve.

Checkup for CHD Patients Who Climb

- **History:** Angina pectoris symptoms in the past? General and specific experience in sports? Medication?
- **Actual status:** Heart, lungs
- **Laboratory:** If applicable, parameters of blood clotting if the patient should take anticoagulants
- **Technical investigations:** ECG, ergometry with analysis of the “ischemic threshold,” echocardiography
- **With individual indication only:** 24 h ECG, stress echocardiography, X-ray thorax, myocardial radio imaging, etc.

Data concerning the necessary exercise tolerance in sport climbing are scarce; most of them originate from very small collectives. Watts et al. recommend from their data encouraging amateur climbers to tolerate a threshold of at least 150/min [18]. Own data indicate a threshold of 120–130/min. (Küpper, unpublished data). All recommendations include the precondition that exercise should not be performed at levels higher than 75–80% of the ischemic threshold [43, 44], which includes a safety margin.

When the climber wants to keep safe, he should respect the following principles. Any symptoms should be stabilized and adequately treated, and exercise tolerance should be sufficient. To monitor the exercise, a pulse watch may be used with its alarm threshold set at 85% of the ischemic threshold. Note: Watches with a fingertip sensor and which measure only if the person puts the finger on this sensor are not acceptable. In general, climbing for fun should be the goal and not “pushing the limits” or competitions. Forced respiration during expulsion should be avoided whenever possible, and slabs or walls should be favored over roofs or overhangs. CHD patients should climb only if they feel completely well and must accept their individual limit of exercise tolerance. This is based on observations which indicate that pulse frequency is correlated with the difficulty of the respective climb [17]. HR is clearly lower on easier overhangs than on more difficult slabs and crimp holds.

According to Gieseler (personal communication), a generally low risk can be assumed when left ventricular function is normal, ergometry fulfills age-adjusted normal values, exercise does not induce ischemia or complex arrhythmia, and relevant stenoses does not exist, but also after a non-complicated myocardial infarction when after 2–3 months climbing will be tried carefully and with slow increase of load.

Be careful if the HR during climbing is often at, or higher than, the ischemic threshold. If exercise-induced arrhythmia should occur, or in cold climate when cold-induced angina is known, the patient should stay for some minutes in moderate cold before he enters the cold zone. Here any exercise should be increased step by step (as the high breathing volume of cold air is the trigger of symptoms!).

No climbing when angina pectoris symptoms should occur at low exercise. When angina is not stable, in the case of severe ventricular dysfunction (ejection fraction <40%), pathological decrease of blood pressure of more than 15 mmHg during exercise, ST segment depression of more than 2 mm while exercising, resuscitation after myocardial infarction with complex arrhythmia less than 6 months ago.

17.4.3 Arrhythmia

Again it must be stated that there are no specific studies for climbers. However, because emergencies caused by arrhythmia are extremely rare in climbing gyms, it can be concluded that the risk should be quite low. A “healthy worker effect” may give additional risk reduction.

Summary

- **Limiting factor:** Maximal load (HR), but not or to a very limited extent maximum muscular force, length of the climb, endurance, altitude, climate, etc.
- **Additional risk factor:** Combination of some specific types of arrhythmia, e.g., in the case of arrhythmia absoluta and a thrombus in the atrium, a sudden increase of HR may mobilize the thrombus and cause a stroke.

- **Common mistake:** Ignorance and missing the diagnosis of occasional “hops” (citation of a high-risk person who later was vitally endangered by Torsade-des-Pointes arrhythmia!)

In sports medicine, the most important question is whether the respective arrhythmia may become life-threatening for the patient (which would mean “no sports until the problem has been solved”). If this is excluded, the next question is whether the arrhythmia may worsen during exercise. Then the limit should be defined as described for CHD (see above) and which should not be exceeded by the patient during sports. Again the pulse watches as described above with alarm set at 85% of the defined threshold should be used.

Checkup for a Climber with Arrhythmia

- **History:** Cardial history (symptoms, collapse, etc.) general and specific sport experience, medication, etc.
- **Actual status:** Symptoms when working or exercising in daily routine?
- **Laboratory:** Normally not necessary, probably electrolytes
- **Technical investigations:** ECG, ergometry, 24 h ECG, echocardiography (especially to detect atrial thrombus in the case of arrhythmia absoluta)
- **Individual indication only:** Stress echocardiography, X-ray thorax, etc.

Concerning exercise tolerance the recommended minimum values were already discussed in the chapter CHD (see above). In analogy one may call it “arrhythmogenic threshold.” Before any advice, the etiology of the respective arrhythmia must be clear because arrhythmia is a symptom and not a diagnosis. It is possible that the main limitation of the person is defined by the underlying disease and not by the risk caused by the arrhythmia.

A low risk may be assumed for climbers when there are some ventricular extrasystoles only which do not cause hemodynamic consequences, and when there are some supraventricular extrasystoles without tendency for tachycardia or in

the case of stable and hemodynamically non-relevant arrhythmia absoluta without thrombus in the atrium [6].

On the other hand, a high risk should be assumed if a pre-excitation syndrome was diagnosed, in the case of intermittent tachycardia independent from its cause and in the case of hemodynamically or complex arrhythmias. Of course, the question whether the patient is fit to climb again may be answered when the therapy has been proven to be sufficient [6].

Having a pacemaker, implanted defibrillator, or an event recorder, is not a general contraindication against climbing or sport in general. Here, the clinical situation is guiding. However, it must be considered that a harness or the belt of a rucksack may apply pressure on the system, and this must be avoided [6]. With modern sport climbing harnesses, the former is not a problem anymore. However, about the latter, any patient should be informed. It is a misbelief that anticoagulant therapy is an absolute contraindication against climbing. Of course any bleeding of minor wounds will be prolonged, or hematoma will be larger. The danger for the patient, however, is intracranial bleeding. Therefore, the patients should wear a helmet without any exception, also indoors in the gym.

17.4.4 Cardiac Insufficiency and Valvular defects

Summary

- **Limiting factor:** Workload, not short maximal muscular forces
- **Additional risk factors:** Climbing at moderate altitude
- **Common mistakes:** Ignorance and overconfidence. Climbing with cardiac insufficiency means climbing for fun, not for “pushing the limits” or competition!

Again the advising physician must live with missing data and studies. However, there is consensus that the climber should strictly avoid massive increases of blood pressure especially on longer athletic routes, routes near or beyond the

Table 17.1 NYHA classification [46], the most commonly used classification of heart failure in Germany since the guidelines were published in 2005 [50]

Class	Definition
1	Heart disease with no limitation to any physical activity. Normal workload and daily routine does not cause undue exhaustion, arrhythmia, shortness of breath, or angina pectoris
2	Heart disease with a mild decrease of the exercise tolerance. No symptoms at rest. Normal daily activities and workload may cause exhaustion, arrhythmia, shortness of breath, or angina pectoris
3	Significant heart disease with marked decrease of exercise tolerance. No symptoms at rest, but easy exercise causes exhaustion, arrhythmia, shortness of breath, or angina pectoris
4	Heart disease with symptoms at rest and during all physical activity, bed confinement

climber’s abilities, or the use of excessive force when climbing technique is lacking. Beyond that any advice orientates mainly on the cardiac exercise capabilities. This will be usually classified according to the New York Heart Association (NYHA) since 1928 (Table 17.1), although it has been modified several times [45, 46]. Again it must be noted that cardiac insufficiency is a symptom and not a diagnosis. The latter must be clear because it may cause the main limitation to the climber and not the insufficiency.

With Stage 1 high end climbing or competition climbing is limited significantly, but motivated recreational climbers will be able to continue their sport without major problems. Some of them may also join recreational competitions. A check by a cardiologist which includes an echocardiography should be performed annually [6].

In Stage 2 competitions and athletic hard routes should be avoided. The limitation is too significant, and an acute problem at such high loads cannot be excluded for sure. Climbing for fun at walls or slabs should be possible without major problems in most cases [6].

In Stage 3 climbing is more or less dominated by social factors. Some easy short and sloped routes may be possible. Belaying is also possible; leading may be critical. Any kind of “pushing the limits” or competition is strictly forbidden.

Stage 4 limits the patient to look and meet others when they are climbing, often with supplemental oxygen. The patient will not be able to climb anymore. This is not a question for sport medicine anymore [6].

Valvular defects are similar to cardiac insufficiency when seen from the viewpoint of exercise physiology. The advice will be therefore guided by the recommendations for cardiac insufficiency [6]. In the case of valvular transplant, the usual infection prophylaxis or consequent therapy of any infection is a must.

17.5 Pulmonary Diseases

Summary

- **Limiting factors:** Whole body load or maximal force, depending on age and type of pulmonary disease. Avoid strictly forced respiration during expulsion when bullae are known or a pneumothorax has happened in the patient’s history! In the case of COPD or silicosis, breathing minute volume or VO_2 may define the limitation.
- **Additional risk factors:** Exercise with active infections in the respiratory system, ignorance of the problem or overconfidence

Again data are scarce. However, there are some studies concerning relevant parameters. In their already referenced papers, the groups of Watts and those of Mermier draw different conclusions from each other regarding the breathing volume and VO_2 of motivated amateur climbers (which means difficulties up to UIAA grade VIII) and increases about three- to fourfold compared to resting values [18, 19]. In competition climbers, Booth et al. (1999) found an six- to eightfold

increase of the respective values. These studies give at least some evidence about the minimal requirements of the lung differentiated according to the sportive intention of the climber.

Checkup for a Climber with Pulmonary Disease

- **History:** Exact diagnosis of the disease, allergies with symptoms of the respiratory system, exercise-dependent symptoms, general and specific sport experience, environmental factors (cold or exercise-induced asthma)
 - **Actual status:** Actual exercise tolerance
 - **Labor:** Red blood cell count, blood gases at rest and during exercise
 - **Technical investigations:** Spiroergometry or, alternatively, ergometry with blood gas analysis at every loading increment, X-ray of the lungs
- With individual indication only:** CT thorax, exclusion of exercise-induced asthma (spirometry before and 5–10 min after maximal exercise)

A low risk may be assumed and sport climbing therefore continued when the symptoms have been therapeutically stabilized and the minimal criteria as described above are fulfilled [6].

A high risk with the consequence that climbing should be stopped at least for a while until the situation has been stabilized, e.g., by drugs. This would be the case if the maximum breathing volume or $\dot{V}O_2$ should be significantly lower than mentioned above. At least a modification of climbing toward less strenuous climbs will be necessary. Beyond this there is a significantly increased risk in the case of large bullae, recurring spontaneous pneumothorax in the patient's history, any acute decompensated or otherwise unstable situation, not or insufficiently treated exercise-induced asthma, cold-induced asthma without special advice how to handle that in the field, relevant hypoxemia when exercising, and severe pulmonary arterial hypertension (systolic >40 mmHg).

Extreme forced respiration during expulsion should be principally avoided, although there are only a few information about incidences and which are all from other sports [47,

48]. Climbing tactics should be adjusted to the sometimes significantly changing status of the patient. For asthma patients or those with COPD, this means that regular peak flow measurements should be performed and documented. With this simple self-test, the patient can monitor and be forewarned of any early worsening in asthma symptoms and adjust medication accordingly before the symptoms become much worse.

17.6 Renal Insufficiency

Summary

- **Limiting factor:** Fluid and electrolyte balance, but not style or intensity of climbing
- **Additional risk factors:** Hot climate/dehydration, catabolic metabolism from training too hard
- **Common mistakes:** Ignorance of fluid or electrolyte balance, climbing during acute infections of the urinary system

Is renal function worsened by climbing? Direct effects are not expected, but indirect ones may occur. The etiology of the underlying condition should be diagnosed and—if possible—treated effectively. In the case of autoimmune diseases, although fortunately rare, an immuno-depressive therapy may be necessary creating limitations for several sports, and because additional infections may further harm the kidneys. The advice to the climber should be to comply with the severity of the disease.

The German classification system for renal insufficiency (Table 17.2) shows two subgroups for Class 3, and neither has any relevance on the advice given to the climber. In stadium 1 and 2, the patient will normally not have any problems when climbing. In stadium 3, the monitoring of fluid balance will be more and more important; in stadium 4, it is a must. If an arteriovenous shunt for dialysis is inserted, it must be protected against any trauma, also minor ones (cover it with soft material). Competition climbing is principally possible in most cases of stadium 1–3 under optimal nephrological coaching. Ideally, the nephrologist is also a climber or has at least some personal experience in climbing.

Table 17.2 Differentiation of chronic renal insufficiency according to ICD-10

ICD-10 code	Description
N18.1	Chronic renal insufficiency, Stadium 1—defect with normal or increased glomerular filtration rate (GFR 90 mL/min or more)
N18.2	Chronic renal insufficiency, Stadium 2—defect with mild decreased glomerular filtration rate (GFR 60–90 mL/min)
N18.3	Chronic renal insufficiency, Stadium 3—defect with moderate decreased glomerular filtration rate (GFR 30–60 mL/min)
N18.4	Chronic renal insufficiency, Stadium 4—defect with significant decreased glomerular filtration rate (GFR 15–30 mL/min)
N18.5	Chronic renal insufficiency, Stadium 5—chronic uremia, terminal renal insufficiency

After a kidney transplantation, the transplant is not well protected somewhere deep in the body but quite superficially situated, and therefore there is a significant risk for injury. Here, the construction and the fitting of the climbing harness are of special importance. Most sit harnesses are constructed in a manner which applies pressure on the region where the transplant is located, especially in the case of a fall [49]. The advice here is to select a harness that fits perfectly on the lower part of the hip that avoids any lifting that might press on the transplant in the event of a fall. The fit should be checked regularly when climbing. If this is observed, there are no other medical contraindications to climb for those with kidney transplant [49].

17.7 Seizures

On the topic “seizures and sport climbing,” there are no systematic studies available. However, there is one international group of experts who has been working intensively about the topic “seizures and mountaineering” [13]. The following statements are mainly based on their work but adapted slightly for sport climbing.

Summary

- **Limiting factors:** Risk caused by acute seizures. During leading, risk of uncontrolled falls. During belaying, complete failure of the

safety chain with the possibility of fatal falls of the leading climber

- **Additional risk factors:** Change of plasma concentrations of anticonvulsive drugs in very hot weather (=dehydration), mistakes in drug intake, and others
- **Common mistakes:** Underestimating of the potential danger when there were no seizures for a longer time

Check of Climbing Persons with Seizures

- **History:** Frequency of attacks? When was the last one? Is there an aura? Medication/compliance?
- **Actual status:** Nothing special
- **Laboratory:** Normally not necessary, if indicated drug monitoring
- **Technical investigations:** Normally not necessary

With this group of diagnoses, the tactics of climbing is of extraordinary importance. When bouldering near the ground, there should be a height which allows the drop and with a spotter behind the climber—this is safe in nearly all situations. Take care that there are no big boulders nearby where the climber might land after a fall! When climbing on “French routes” where there is a bolt every 1 or 2 m, it is also safe for the patient to lead. Realistically, these are not real leads but more of a “self-established top-rope” as two bolts can be clipped from the same position and are in front of the climber’s body. In the case of a seizure during the lead, the belayer should let the climber down with controlled speed, while—if possible—another climber cares and protects the head of the victim.

The advice differs once the distance between bolts becomes longer. Here grounders (falls where the climber lands on the ground instead of in the rope/harness) are a special danger for persons with seizures. When sport climbing outdoors on rocks, the situation again is completely different. Here, the distances between the bolts are usually longer, and falls may be long. In such situations, leading is not recommended for persons with seizures. If abseiling should be necessary, it is

possible for these patients; however, they should be belayed with another rope or by a companion who may put load on the rope to block the brake in the case of a problem of the patient.

The role of the belayer if he should suffer from a seizure is completely different as this may cause a complete failure of the safety chain. If seizure patients belay others, they should at least use fully automatic belaying systems (e.g., grigri).

The important question is how long an increased risk should be assumed and when limitations may be eased. In contrast to urban traffic, this has not been regulated by legislation. However, traffic regulations may usefully serve as a reference model since there are also dangerous situations that may arise if a person should become incapacitated when driving. Driving regulations differentiate several situations but also take into account that probably a car may put several persons at risk which is not the case in climbing. Therefore, a seizure-free period of at least 1 year may be used as benchmark to resume leading. In some cases, it may be possible to reduce this to 6 months when a specialist advises the patient and no seizures have occurred without drugs. For alpine climbing, the situation should be handled more strictly. Here, at least 2 years without seizures should be recommended.

It must be pointed out that these are consensus recommendations of specialists from 35 countries (UIAA Medical Commission) which were put together in all conscience from non-systematic studies, observations from practice and case reports. They are not evidence-based. It may be necessary to modify them if new studies should be available. The most recent version will be always found on the website of UIAA MedCom at www.theuiaa.org/medical_advise.html.

17.8 Conclusion

Most pre-existing diseases do not mean “stop climbing!” However, a detailed and specific assessment of the patient is a must, requiring a specific competence. It is advantageous if the advising physician is also a climber as many may not be aware of climbing’s unique physiology and

thus unnecessarily advise terminating all climbing participation, when all that may be needed is a slight modification to climbing participation. In this way, it is possible to retain the ongoing physical and mental health benefits of sport climbing for the patient, which also improves the symptoms and outcome of the disease itself.

References

1. Liguori G. In: (ACSM) ACoSM, editor. ACSM’s guidelines for exercise testing and prescription. 11th ed. Wolters Kluwer Health, Inc.; 2021.
2. Zhang Y, Xu L, Yao Y, Guo X, Sun Y, Zhang J, et al. Effect of short-term exercise intervention on cardiovascular functions and quality of life of chronic heart failure patients: a meta-analysis. *J Exerc Sci Fit*. 2016;14(2):67–75.
3. Pescatello LS, MacDonald HV, Lamberti L, Johnson BT. Exercise for hypertension: a prescription update integrating existing recommendations with emerging research. *Curr Hypertens Rep*. 2015;17(11):87.
4. Engbert K, Weber M. The effects of therapeutic climbing in patients with chronic low back pain: a randomized controlled study. *Spine*. 2011;36(11):842–9.
5. Karg N, Dorscht L, Kornhuber J, Luttenberger K. Boulderer psychotherapy is more effective in the treatment of depression than physical exercise alone: results of a multicentre randomised controlled intervention study. *BMC Psychiatry*. 2020;20(1):116.
6. Küpper T, Morrison A, Gieseler U, Schöffl V. sport climbing with pre-existing cardio-pulmonary medical conditions. *Int J Sports Med*. 2009;30:395–402.
7. Sheel AW, Seddon N, Knight A, McKenzie DC, DE RW. Physiological responses to indoor rock-climbing and their relationship to maximal cycle ergometry. *Med Sci Sports Exerc*. 2003;35(7):1225–31.
8. Sheel AW. Physiology of sport rock climbing. *Br J Sports Med*. 2004;38(3):355–9.
9. Scherer D, Kaltenbach M. [Frequency of life-threatening complications associated with exercise testing (author’s transl)]. *Dtsch Med Wochenschr*. 1979;104(33):1161–1165.
10. Gibbons L, Blair SN, Kohl HW, Cooper K. The safety of maximal exercise testing. *Circulation*. 1989;80(4):846–52.
11. Fletcher GF, Ades PA, Kligfield P, Arena R, Balady GJ, Bittner VA, et al. Exercise standards for testing and training: a scientific statement from the American Heart Association. *Circulation*. 2013;128(8):873–934.
12. Milledge J, Küpper T. Consensus Statement of the UIAA Medical Commission Vol. 13: People with pre-existing conditions going to the mountains. Bern/ Switzerland: The Medical Commission of the Union Internationale des Associations d’Alpinisme (UIAA); 2008. www.theuiaa.org/medical_advice.html.

13. Angelini C, Giardini G. Consensus Statement of the UIAA Medical Commission Vol. 16: Travel to altitude with neurological disorders. Bern/Switzerland: The Medical Commission of the Union Internationale des Associations d'Alpinisme (UIAA); 2009.
14. Donegani E, Hillebrandt D, Windsor J, Küpper T, Gieseler U, Rodway GW. Consensus statement of the UIAA Medical Commission Vol. 21: People with pre-existing cardiovascular conditions going to the mountains. Bern/Switzerland: Union Internationale des Associations d'Alpinisme, Medical Commission; 2012. www.theuiaa.org/medical_advice.html.
15. Donegani E, Hillebrandt D, Windsor J, Gieseler U, Rodway GW, Schöffl V, et al. People with pre-existing cardiovascular conditions going to the mountains consensus statement of the medical commission of the Union Internationale des Associations d'Alpinisme (UIAA MedCom). *Travel Med Inf Dis (TMAID)*. 2014;12(3):237–52.
16. Hillebrandt D, Gurtoo A, Küpper T, Richards P, Schöffl V, Shah P, et al. UIAA medical commission recommendations for mountaineers, hillwalkers, trekkers, and rock and ice climbers with diabetes. *High Alt Med Biol*. 2018.
17. Watts PB. Physiology of difficult rock climbing. *Eur J Appl Physiol*. 2004;91(4):361–72.
18. Watts PB, Daggett M, Gallagher P, Wilkins B. Metabolic response during sport rock climbing and the effects of active versus passive recovery. *Int J Sports Med*. 2000;21(3):185–90.
19. Mermier CM, Robergs RA, McMinn SM, Heyward VH. Energy expenditure and physiological responses during indoor rock climbing. *Br J Sports Med*. 1997;31(3):224–8.
20. Williams ES, Taggart P, Carruthers M. Rock climbing: observations on heart rate and plasma catecholamine concentrations and the influence of oxprenolol. *Br J Sports Med*. 1978;12(3):125–8.
21. Mermier CM, Janot JM, Parker DL, Swan JG. Physiological and anthropometric determinants of sport climbing performance. *Br J Sports Med*. 2000;34(5):359–65; discussion 66.
22. Booth J, Marino F, Hill C, Gwinn T. Energy cost of sport rock climbing in elite performers. *Br J Sports Med*. 1999;33(1):14–8.
23. Limonta E, Brighenti A, Rampichini S, Ce E, Schena F, Esposito F. Cardiovascular and metabolic responses during indoor climbing and laboratory cycling exercise in advanced and elite climbers. *Eur J Appl Physiol*. 2018;118(2):371–9.
24. Maron BJ, Levine BD, Washington RL, Baggish AL, Kovacs RJ, Maron MS. Eligibility and disqualification recommendations for competitive athletes with cardiovascular abnormalities: task force 2: preparticipation screening for cardiovascular disease in competitive athletes: a scientific statement from the American Heart Association and American College of Cardiology. *J Am Coll Cardiol*. 2015;66(21):2356–61.
25. Boden BP, Breit I, Beachler JA, Williams A, Mueller FO. Fatalities in high school and college football players. *Am J Sports Med*. 2013;41(5):1108–16.
26. Gouttebarga V, Ooms W, Tummers T, Inklaar H. Mortality in international professional football (soccer): a descriptive study. *J Sports Med Phys Fitness*. 2015;55(11):1376–82.
27. Morentin B, Suarez-Mier MP, Monzo A, Molina P, Lucena JS. Sports-related sudden cardiac death due to myocardial diseases on a population from 1–35 years: a multicentre forensic study in Spain. *Forensic Sci Res*. 2019;4(3):257–66.
28. Roberts JA. Viral illnesses and sports performance. *Sports Med (Auckland, NZ)*. 1986;3(4):298–303.
29. Breinholt JP, Moulik M, Dreyer WJ, Denfield SW, Kim JJ, Jefferies JL, et al. Viral epidemiologic shift in inflammatory heart disease: the increasing involvement of parvovirus B19 in the myocardium of pediatric cardiac transplant patients. *J Heart Lung Transplant*. 2010;29(7):739–46.
30. Kindermann I, Ukena C, Mahfoud F, Böhm M, Yilmaz A, Klingel K. Myokarditis-update. *Kardiologie*. 2016;10(5):311–30.
31. Yilmaz A, Klingel K, Kandolf R, Sechtem U. A geographical mystery: do cardiotropic viruses respect national borders? *J Am Coll Cardiol*. 2008;52(1):82; author reply-3.
32. Küpper T, Schraut B, Rieke B, Schöffl V, Steffgen J. Drugs and drug administration in extreme climates. *J Travel Med*. 2006;13(1):35–47.
33. Löllgen H. Bedeutung und Evidenz der körperlichen Aktivität zur Prävention und Therapie von Erkrankungen. *Dtsch Med Wochenschr*. 2013;138(1):1–8.
34. Löllgen H, Dickhuth HH, Dirschedl P. Vorbeugung durch körperliche Bewegung. *Dt Ärztebl*. 1998;95(24):B1228–B34.
35. Löllgen H, Gerke RJ, Steinberg T. Der kardiale Zwischenfall beim Sport. *Dt Ärztebl*. 2006;103(23):B1382–B7.
36. Küpper T. Sportklettern mit Hypertonie. In: Kupper T, Ebel K, Gieseler U, editors. *Moderne Berg- und Höhenmedizin*. 2nd. ed. Stuttgart: Gentner Verlag; 2011. p. 563–5.
37. Aras D, Akalan C, Koz M, Ileri M. Does long term sport rock climbing training affect on echocardiography and heart rate variability in sedentary adults? A randomized, and controlled study. *IJAEP*. 2016;5(5):26–35.
38. Ferguson RA, Brown MD. Arterial blood pressure and forearm vascular conductance responses to sustained and rhythmic isometric exercise and arterial occlusion in trained rock climbers and untrained sedentary subjects. *Eur J Appl Physiol Occup Physiol*. 1997;76(2):174–80.
39. Samek L, Löllgen H. Belastungs-EKG: Ischämiediagnostik. In: Lollgen H, Winter HJ, Erdmann E, editors. *Ergometrie—Belastungsuntersuchungen in Klinik und Praxis*. Berlin/Heidelberg: Springer Verlag; 1995. p. 111–26.

40. Löllgen H. Kardiopulmonale Funktionsdiagnostik [Cardiopulmonary function diagnosis]. 2. Aufl. ed. Basel: Ciba; 1990.
41. Löllgen H, Winter UJ. Ergometrien in der Prävention—Reihenuntersuchungen. In: Löllgen H, Winter UJ, Erdmann E, editors. Ergometrie—Belastungsuntersuchungen in Klinik und Praxis. Berlin/Heidelberg: Springer Verlag; 1995. p. 319–22.
42. Limb D. Injuries on British climbing walls. *Br J Sports Med.* 1995;29(3):168–70.
43. Hollmann W, Hettinger T. Sportmedizin, Grundlagen für Arbeit, Training und Präventivmedizin. Stuttgart: Schattauer; 2000.
44. Morgan BJ, Alexander JK, Nicoli SA, Brammell HL. The patient with coronary heart disease at altitude: observations during acute exposure to 3100 meters. *J Wilderness Med.* 1990;1:147–53.
45. (anonymous). The criteria committee of the New York Heart Association. Diseases of the heart and blood vessels: nomenclature and criteria for diagnosis. In: N.N., editor. The Criteria Committee of the New York Heart Association diseases of the heart and blood vessels: nomenclature and criteria for diagnosis. 6th. ed. Boston, MA: Little Brown; 1928.
46. (anonymous). The Criteria Committee of the New York Heart Association. Nomenclature and Criteria for Diagnosis of Diseases of the Heart and Great Vessels. In: N.N., editor. The Criteria Committee of the New York Heart Association Nomenclature and Criteria for Diagnosis of Diseases of the Heart and Great Vessels. 9th ed. ed. Boston, Mass.: Little, Brown & Co; 1994. p. 253–6.
47. Sadarangani S, Patel DR, Pejka S. Spontaneous pneumomediastinum and epidural pneumatosis in an adolescent precipitated by weight lifting: a case report and review. *Phys Sportsmed.* 2009;37(4):147–53.
48. Davis PF. Primary spontaneous pneumothorax in a track athlete. *Clin J Sport Med.* 2002;12(5):318–9.
49. Welch TR. Climbing harness fit in kidney transplant recipients. *Wilderness Environ Med.* 1999;10(1):3–5.
50. Hoppe UC, Böhm M, Dietz R, Hanrath P, Kroemer HK, Osterspey A, et al. Leitlinien zur Therapie der chronischen Herzinsuffizienz (2005). *Z Kardiol.* 2005;94:488–509.